

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458886.8926 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.032 +/- 0.035
Transit depth (Rp/Rs)^2	0.1 +/- 0.22 [%]
Orbital Inclination (inc)	80.83 +/- 3.0
Airmass coefficient 1 (a1)	1.186 +/- 0.011
Airmass coefficient 2 (a2)	-0.121 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	4.14
Optimal Annulus	13.11
Transit Duration (day)	0.034 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.032 ± 0.035 vs 0.1594 (deviation 3.52σ)
Duration fit vs published	0.034 d vs 0.0512 d (deviation 1.04σ)
Mid-time O–C	4.2 min
Depth SNR	0.9
Residual scatter	0.88 %

Light curve

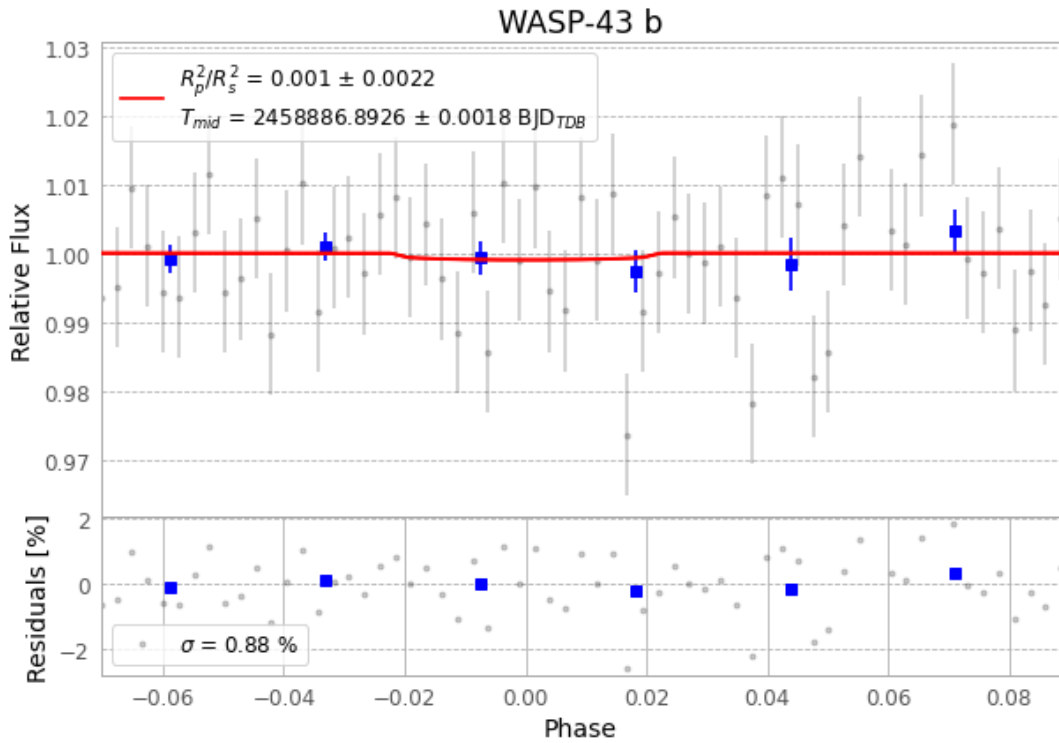


Figure 1 — FinalLightCurve_WASP-43 b_2020-02-07.png (SHA-256 c96d23884d40c450...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [238, 393], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d14061a27321eb64...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), WASP-43200207075412.FITS → WASP-43200207110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200207.log (3956880e74cc...), FinalLightCurve_WASP-43 b_2020-02-07.png (c96d23884d40...), FinalLightCurve_WASP-43 b_2020-02-07.csv (249b2dddae30...)

Compute resources

Wall time 136.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 14:57Z.